For Examiner's Use

4 **(a)** Explain,

(i) what is meant by a *radian*,

.....

.....

.....[2]

(ii) why one complete revolution is equivalent to an angular displacement of 2π rad.

.....

.....[1]

(b) A stone of weight 3.0 N is fixed, using glue, to one end P of a rigid rod CP, as shown in Fig. 4.1.

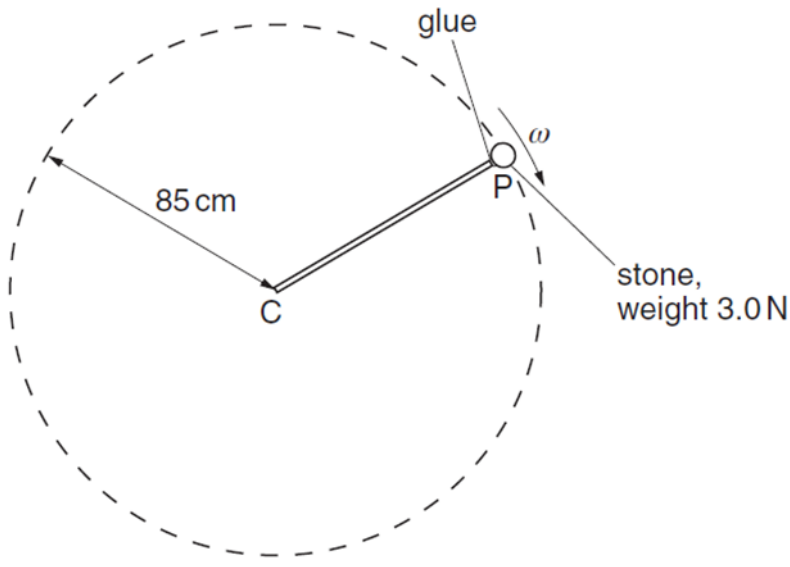

Fig. 4.1

The rod is rotated about end C so that the stone moves in a vertical circle of radius 85 cm.

The angular speed ω of the rod and stone is gradually increased from zero until the glue snaps. The glue fixing the stone snaps when the tension in it is 18 N.

For Examiner's Use

For the position of the stone at which the glue snaps,

(i) mark with the letter S, the position of the stone on the dotted circle of Fig. 4.1 and [1]

(ii) calculate the angular speed ω of the stone.

angular speed = ... rad s^{-1} [3]

(c) The same stone is now fixed on a string and made to travel along a horizontal circular path, as shown in Fig. 4.2.

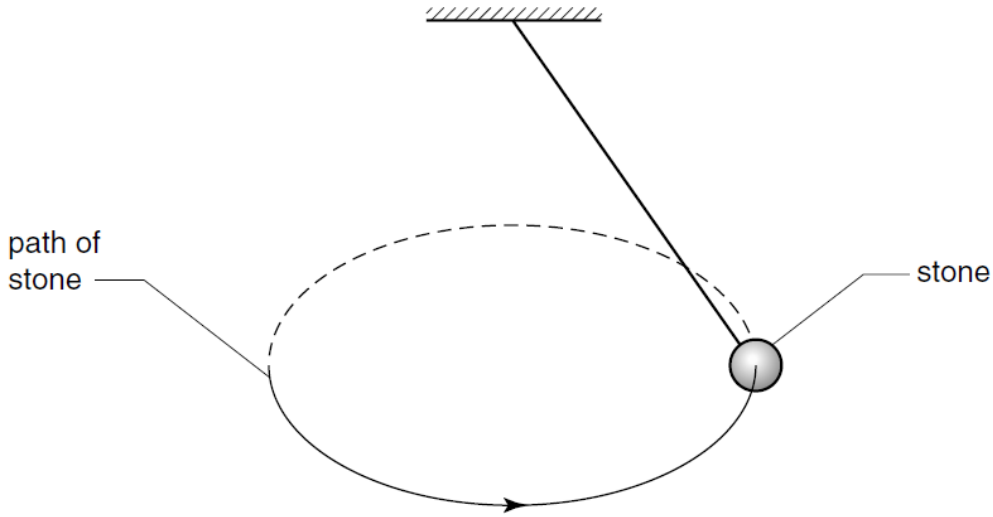

Fig. 4.2

The string makes an angle of 35° to the vertical, as illustrated in Fig. 4.3.

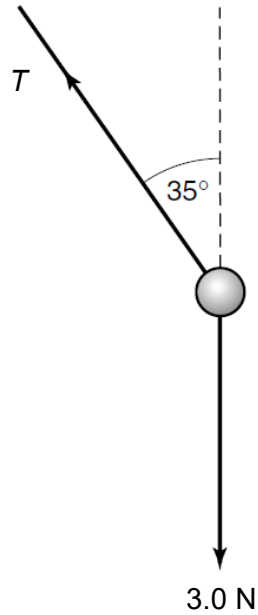

Fig. 4.3

Determine

(i) the tension *T* in the string, and

tension = ... N [1]

(ii) the resultant force acting on the stone in the position shown.

magnitude of force = ... N

direction of force = ... [2]